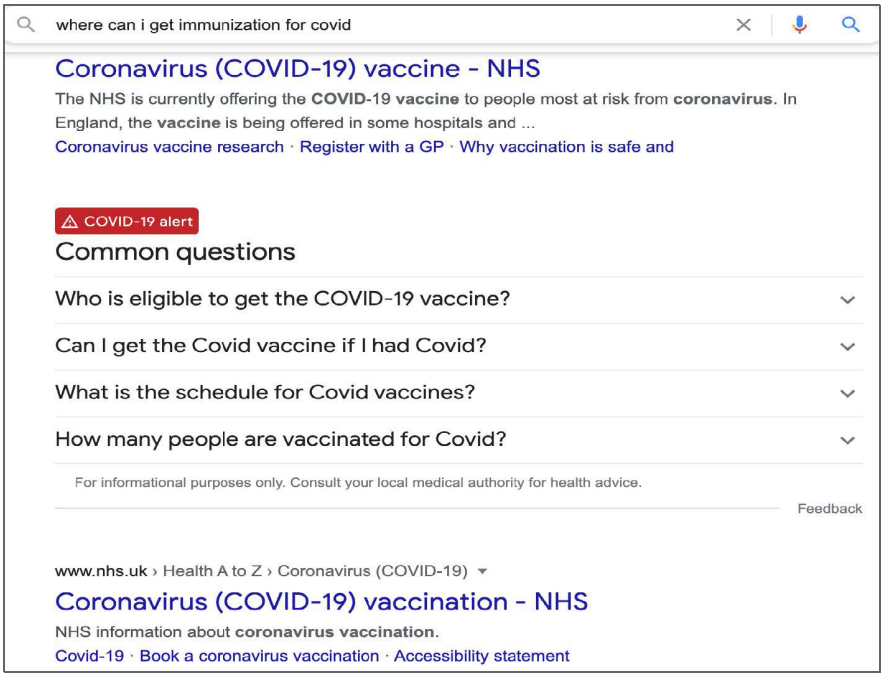


Figure 3.2 Search results (as expected) from Google

We learn about this analysis phase in chapter 7. Once data has been analyzed, it is sent to a particular data node for persistence. Usually, text analysis, persistence activities, data replication, and so on are carried out in a fraction of a second, so the data is ready for consumption in less than a second after it is indexed (a service level agreement [SLA] that Elasticsearch vouches for). The next step after data processing in Elasticsearch is data retrieval in the form of search and analytics, which we discuss in the next section.

3.1.3 Data out

After data is collected, analyzed, and stored in Elasticsearch, it is retrieved via search and analytical queries. Search fetches matching data for a specified query, while analytics gathers data to form summarized statistics. Search looks not just for exact word-to-word matches but also for root words, synonyms, spelling mistakes, and more. The advanced features of full-text queries are what make modern search engines like Elasticsearch as preferred choices for search.

When a search query is issued, if the field is a full-text field, it undergoes an analysis phase similar to that performed when the field was indexed. That is, the query is tokenized and normalized using the same analyzers associated with that field. The respective tokens are searched and matched in the inverted index, and results based on the matches are relayed back to the client. For example, if a field is set with a French